

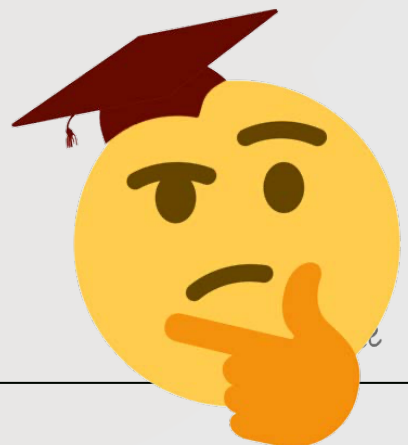
10 – (MathAA-S 2021-Paper 2/2-Q7) – *PERMUTATION - COMBINATION*

[Maximum mark: 5]

Eight runners compete in a race where there are no tied finishes. Andrea and Jack are two of the eight competitors in this race.

Find the total number of possible ways in which the eight runners can finish if Jack finishes

- (a) in the position immediately after Andrea; [2]
- (b) in any position after Andrea. [3]



1 – (MAT-S 2012-Paper 2/1-Q9) – BINOMIAL THEOREM

[Maximum mark: 7]

Find the constant term in the expansion of $\left(x - \frac{2}{x}\right)^4 \left(x^2 + \frac{2}{x}\right)^3$.

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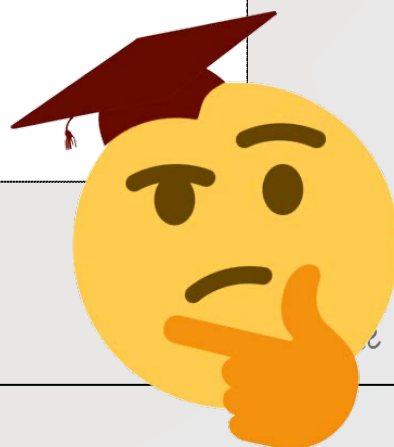
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2 – (MAT-S 2014-Paper 2/2-Q5) – BINOMIAL THEOREM*[Maximum mark: 6]*

Find the coefficient of x^{-2} in the expansion of $(x-1)^3 \left(\frac{1}{x} + 2x \right)^6$.

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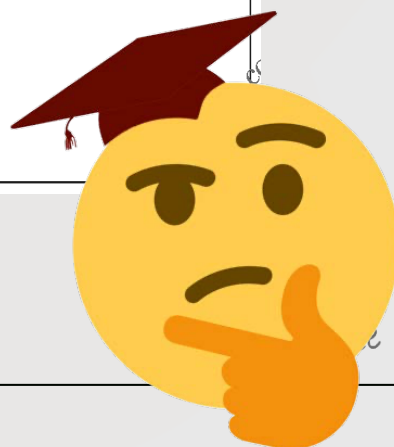
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3 – (MAT-S 2014-Paper 2/1-Q8) – *BINOMIAL THEOREM, PROBABILITY*

- (a) Find the term in x^5 in the expansion of $(3x + A)(2x + B)^6$. [4]

Mina and Norbert each have a fair cubical die with faces labelled 1, 2, 3, 4, 5 and 6; they throw it to decide if they are going to eat a cookie.

Mina throws her die just once and she eats a cookie if she throws a four, a five or a six. Norbert throws his die six times and each time eats a cookie if he throws a five or a six.

- (b) Calculate the probability that five cookies are eaten. [4]

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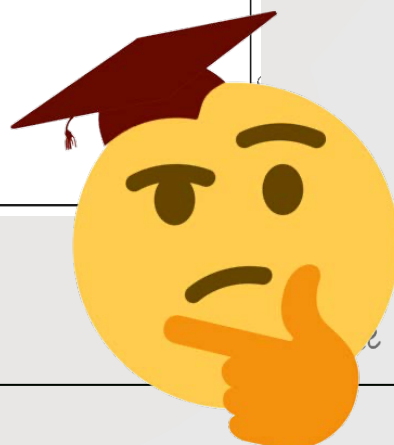
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4 – (MAT-W 2016-Paper 2/0-Q4) – *BINOMIAL THEOREM*

[Maximum mark: 5]

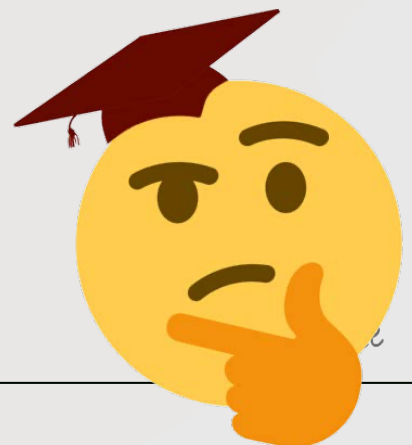
Find the constant term in the expansion of $\left(4x^2 - \frac{3}{2x}\right)^{12}$.



5 – (MAT-S 2017-Paper 2/1-Q3) – *BINOMIAL THEOREM*

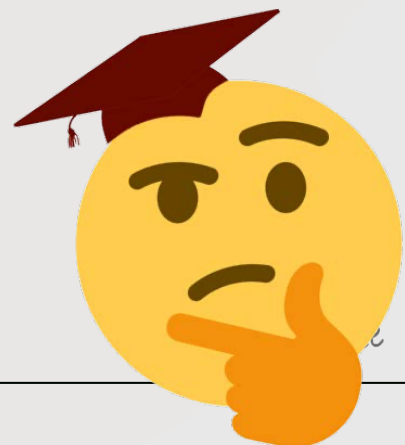
[Maximum mark: 6]

The coefficient of x^2 in the expansion of $\left(\frac{1}{x} + 5x\right)^8$ is equal to the coefficient of x^4 in the expansion of $(a + 5x)^7$, $a \in \mathbb{R}$. Find the value of a .



6 – (MAT-S 2018-Paper 2/2-Q5) – *BINOMIAL THEOREM*

- (a) Express the binomial coefficient $\binom{3n+1}{3n-2}$ as a polynomial in n . [3]
- (b) Hence find the least value of n for which $\binom{3n+1}{3n-2} > 10^6$. [3]



7 – (MAT-W 2018-Paper 2/0-Q4) – *BINOMIAL THEOREM*

[Maximum mark: 5]

Find the value of the constant term in the expansion of $x^4 \left(x + \frac{3}{x^2} \right)^5$.



8 – (MAT-W 2019-Paper 2/0-Q5) – *BINOMIAL THEOREM*

[Maximum mark: 6]

Consider the expansion of $(2 + x)^n$, where $n \geq 3$ and $n \in \mathbb{Z}$.

The coefficient of x^3 is four times the coefficient of x^2 . Find the value of n .

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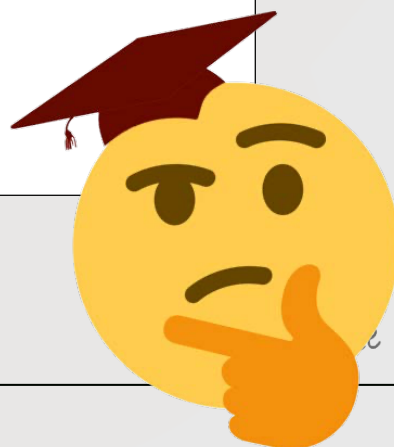
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9 – (MAT-W 2020-Paper 2/0-Q4) – *BINOMIAL THEOREM*

[Maximum mark: 6]

Find the term independent of x in the expansion of $\frac{1}{x^3} \left(\frac{1}{3x^2} - \frac{x}{2} \right)^9$.



10 – (MathAA-S 2021-Paper 2/1-Q5) – *BINOMIAL THEOREM*

[Maximum mark: 5]

Consider the expansion of $(3 + x^2)^{n+1}$, where $n \in \mathbb{Z}^+$.

Given that the coefficient of x^4 is 20 412, find the value of n .

